

James Kim

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Full-stack and AI product engineer. I build the interfaces people actually touch, and lately that has meant putting AI inside them. React, TypeScript, Python, GraphQL.

EXPERIENCE

Ironflow AI - Software Engineer Intern

Solana Beach, CA | June 2026 - Sept 2026

Multi-tenant ERP for manufacturers. React 19, Next.js, TypeScript, Python, FastAPI, GraphQL, PostgreSQL, Docker.

- Shipped features and fixes across the whole stack, from CI and infrastructure through the backend to frontend product work, owning each from ticket through code review to a staged, feature-flagged rollout, at a sustained pace, every one QA-verified before release.
- Cut the AI assistant's answer formatting delay from roughly **9 seconds to zero** by batching server-sent-event chunks into a 50 ms buffer, memoizing the React render path and adding an adaptive typewriter reveal, so long answers format while they stream rather than after.
- Built the AI agent's clarifying-question flow end to end, so it asks rather than guessing on an ambiguous request: an agent tool on the Anthropic API and Strands Agents, a new event threaded through the server-sent-event pipeline, and one React component answering single-select, multi-select and free-text with a skip path, covered by Storybook states and shipped behind a feature flag defaulted off.
- Delivered a multi-level bill-of-materials view end to end, database through API to UI, taking it from a single level of components to every level, by modelling the bill as a graph rather than a tree so a sub-assembly shared by several parents renders correctly under each.
- Made the product usable for colorblind operators with themes built on the Okabe-Ito palette, remapping only the tokens that carry meaning through hue.
- Built self-service API token management end to end: a GraphQL mutation that derives identity from the auth context so a caller can only mint their own token, stores a hash and returns the secret once, plus the settings UI to mint and revoke it.
- Wrote the automated test coverage that shipped with all of it, in Jest, React Testing Library and pytest, reviewed teammate pull requests in depth, and improved the reliability of the continuous integration pipeline.

PROJECTS

FantasyStats: Odds, Not Projections | TypeScript, Monte Carlo, WebGL, Cloudflare

2026

- Measured across four seasons of play that a trained model does not beat a simple season average, so the product is the uncertainty: every player is a simulated distribution, the headline is the chance of a big week, and two players whose distributions overlap are reported as tied rather than ranked.
- Monte Carlo simulation with the spread drawn from a position-and-mean model rather than from the player, because week-to-week volatility turns out to be mostly a function of how good someone is. Walk-forward backtest with leakage blocked by the type system, and the page publishes its own scorecard against what happened.

sidereal: Real-Time Shared Sky | Cloudflare Workers, Durable Objects, WebGL2, TypeScript

2026

- A night sky anyone can open and look around together, where every light traces to a live measurement: Wikipedia edits, USGS earthquakes, the ISS and the NOAA aurora forecast, pushed to every visitor over one WebSocket.
- Draws **8,920 Hipparcos stars** and the live events in one instanced WebGL2 pass at under 1 ms a frame, brightness on the real logarithmic magnitude scale. **545 tests**, WCAG AA verified by an automated probe across 15 palettes, and a provenance audit that fails the build if anything is drawn without a measurement behind it.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java, Go, Bash

Frontend: React 19, Next.js, Apollo Client, GraphQL, Tailwind CSS, Jest, React Testing Library, Storybook, WebGL, responsive UI, accessibility (WCAG)

Backend: FastAPI, Strawberry GraphQL, REST, SQLAlchemy, Alembic, PostgreSQL, MySQL, Pydantic, pytest, Cloudflare Workers, WebSockets

AI: AI agents, large language models (LLMs), generative AI, Anthropic API, Strands Agents, Model Context Protocol (MCP), Gemini API, prompt engineering, streaming (SSE)

Practices: Git, GitHub Actions, CI/CD, code review, unit and integration testing, Docker, feature flags (LaunchDarkly), Agile

EDUCATION

University of Mississippi - B.A. Computer Science, Minor in General Business

Oxford, MS | Aug 2022 - May 2026

LEADERSHIP AND HONORS

- President, Upsilon Pi Epsilon (UPE) Honor Society**, Aug 2025 - May 2026. Led the Ole Miss chapter of the international computing honor society and ran the full member induction process.
- HackBeta 2024: 1st Place Overall** and 1st, Challenge Category, for a cross-platform Java file-transfer system. **HackBeta 2025: 2nd Place.**